REPRESENTABILITY OF THE DIAGONAL OF A PRESHEAF VIA FIBRE PRODUCTS OF REPRESENTABLES

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Editorial note. Counted span is the author's own proof environment, source lines 1232-1278, following the statement at 1216-1230.

Editorial note. The chapter's Yoneda input is used as a framework fact; the presheaf fibre product description is the in-chapter lemma referenced at 1257-1259.

Allowed standard prerequisites (editorial).

- Yoneda identification. Yoneda lemma of this chapter, used at proof lines 1233-1234 and 1266-1268 (framework input; the chapter's Yoneda statement carries bibliographic citations and is not itself carded). Identifies F(U) with transformations h_U -> F and turns projections into morphisms in C.
- Explicit fibre products of presheaves. Lemma lemma-fibre-product-presheaves of this chapter, invoked at proof lines 1257-1259. The explicit description lets the reader read off h_W = h_U x_{xi,F,xi'} h_V from the cartesian square.

Source reference. Original lemma 0022: sources/stacks-categories/pinned-original-source.tex, source lines 1157–1157. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{C}$ be a category. Let $F : \mathcal{C}^{opp} \to Sets$ be a functor. Assume $\mathcal{C}$ has products of pairs of objects and fibre products. The following are equivalent:*

1. *the diagonal $\Delta : F \to F \times F$ is representable,*
2. *for every U in $\mathcal{C}$, and any $\xi \in F(U)$ the map $\xi : h_U \to F$ is representable,*
3. *for every pair U, V in $\mathcal{C}$ and any $\xi \in F(U)$, $\xi' \in F(V)$ the fibre product $h_U \times_{\xi,F,\xi'} h_V$ is representable.*

Proof. We will continue to use the Yoneda lemma to identify $F(U)$ with transformations $h_U \to F$ of functors.

Equivalence of (2) and (3). Let U, ξ, V, ξ' be as in (3). Both (2) and (3) tell us exactly that $h_U \times_{\xi,F,\xi'} h_V$ is representable; the only difference is that the statement (3) is symmetric in U and V whereas (2) is not.

Assume condition (1). Let U, ξ, V, ξ' be as in (3). Note that $h_U \times h_V = h_{U \times V}$ is representable. Denote $\eta : h_{U \times V} \to F \times F$ the map corresponding to the product $\xi \times \xi' : h_U \times h_V \to F \times F$. Then the fibre product $F \times_{\Delta,F \times F,\eta} h_{U \times V}$ is representable

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by assumption. This means there exist $W \in \mathrm{Ob}(\mathcal{C})$, morphisms $W \to U$, $W \to V$ and $h_W \to F$ such that

$$\begin{array}{ccc} h_W & \longrightarrow & h_U \times h_V \\ \downarrow & & \downarrow{\scriptstyle \xi \times \xi'} \\ F & \longrightarrow & F \times F \end{array}$$

is cartesian. Using the explicit description of fibre products in Lemma 0022 the reader sees that this implies that $h_W = h_U \times_{\xi, F, \xi'} h_V$ as desired.

Assume the equivalent conditions (2) and (3). Let U be an object of $\mathcal{C}$ and let $(\xi, \xi') \in (F \times F)(U)$. By (3) the fibre product $h_U \times_{\xi, F, \xi'} h_U$ is representable. Choose an object W and an isomorphism $h_W \to h_U \times_{\xi, F, \xi'} h_U$. The two projections $\mathrm{pr}_i : h_U \times_{\xi, F, \xi'} h_U \to h_U$ correspond to morphisms $p_i : W \to U$ by Yoneda. Consider $W' = W \times_{(p_1, p_2), U \times U} U$. It is formal to show that W' represents $F \times_{\Delta, F \times F} h_U$ because

$$h_{W'} = h_W \times_{h_U \times h_U} h_U = (h_U \times_{\xi, F, \xi'} h_U) \times_{h_U \times h_U} h_U = F \times_{F \times F} h_U.$$

Thus Δ is representable and this finishes the proof. $\square$